Cardiovexaban in High-Risk Patients with Atherosclerotic Cardiovascular Disease

The SHIELD-1 Trial

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ABSTRACT

Background

Despite guideline-directed medical therapy (GDMT), many patients with established atherosclerotic cardiovascular disease (ASCVD) remain at elevated risk for recurrent major adverse cardiovascular and cerebrovascular events (MACCE). Cardiovexaban is a novel dual-pathway antiplatelet agent that inhibits both P2Y12 and protease-activated receptor-1 (PAR-1), providing deeper platelet inhibition than single-pathway agents.

Methods

In this multicenter, randomized, double-blind, placebo-controlled trial, we enrolled 8,206 patients with documented ASCVD (prior myocardial infarction, ischemic stroke, or peripheral arterial disease) and elevated residual cardiovascular risk (10-year ASCVD risk score $\geq$15% or $\geq$2 additional high-risk features) despite GDMT. Patients were randomly assigned in a 1:1 ratio to receive cardiovexaban 150 mg once daily or matching placebo, in addition to background antiplatelet and statin therapy. The primary efficacy endpoint was time to first occurrence of the composite of cardiovascular death, myocardial infarction, stroke, or urgent coronary revascularization (four-point MACCE). The primary safety endpoint was TIMI major bleeding.

Results

Over a median follow-up of 36 months, the primary endpoint occurred in 489 patients (11.9%) in the cardiovexaban group and 643 patients (15.7%) in the placebo group (hazard ratio, 0.75; 95% confidence interval [CI], 0.64 to 0.88; $P<0.001$). TIMI major bleeding occurred in 156 patients (3.8%) receiving cardiovexaban versus 86 patients (2.1%) receiving placebo (hazard ratio, 1.82; 95% CI, 1.39 to 2.38; $P<0.001$). Most bleeding events were gastrointestinal (70%); intracranial hemorrhage rates were low and balanced between groups (0.4% vs. 0.3%).

Conclusions

Among high-risk patients with established ASCVD on GDMT, cardiovexaban significantly reduced the risk of MACCE compared with placebo, with an increase in major bleeding events driven primarily by gastrointestinal bleeding. The benefit-risk profile supports use in selected high-risk secondary prevention populations.

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1 Introduction

Atherosclerotic cardiovascular disease (ASCVD) remains the leading cause of morbidity and mortality worldwide, accounting for more than 18 million deaths annually. Despite advances in lipid-lowering therapy, blood pressure control, and antiplatelet treatment, patients with established ASCVD continue to experience recurrent cardiovascular events at substantial rates. Current guidelines recommend aspirin or a P2Y12 inhibitor (clopidogrel, prasugrel, or ticagrelor) in combination with high-intensity statin therapy for secondary prevention. However, even with optimal guideline-directed medical therapy (GDMT), 10-year event rates for major adverse cardiovascular and cerebrovascular events (MACCE) remain 12% to 20% in contemporary registries, indicating significant residual thrombotic risk.

The concept of residual risk reflects the observation that standard antiplatelet therapy incompletely suppresses platelet activation through multiple redundant signaling pathways. While aspirin inhibits thromboxane A2-mediated platelet activation and P2Y12 inhibitors block ADP-mediated activation, other pathways—most notably thrombin-mediated activation through protease-activated receptor-1 (PAR-1)—remain intact. PAR-1 is the primary thrombin receptor on human platelets and a major driver of thrombotic events in the setting of atherosclerotic plaque rupture. Preclinical studies have demonstrated that dual inhibition of P2Y12 and PAR-1 provides more complete suppression of platelet aggregation than P2Y12 inhibition alone.

Cardiovexaban is a novel oral antiplatelet agent that selectively inhibits both the P2Y12 receptor (similar to clopidogrel and ticagrelor) and the PAR-1 receptor. In phase 2 dose-ranging studies, cardiovexaban 150 mg once daily demonstrated superior platelet inhibition compared with clopidogrel monotherapy, with pharmacodynamic effects sustained over 24 hours. Early safety signals showed increased bleeding compared with single-pathway inhibition, but with most events being minor or moderate in severity.

The Safety and efficacy of cardiovexaban for High-rIsk sEcondary prevention of atheroschLerotic Disease (SHIELD-1) trial was designed to evaluate whether dual-pathway platelet inhibition with cardiovexaban reduces MACCE in high-risk patients with established ASCVD already receiving GDMT, and to characterize the bleeding risk associated with this approach.

2 Methods

2.1 Trial Design and Oversight

SHIELD-1 was a multicenter, randomized, double-blind, placebo-controlled, event-driven trial conducted at 412 sites in 28 countries across North America, Europe, Asia-Pacific, and Latin America. The trial was designed by the academic steering committee in collaboration with the sponsor (Novellia Pharmaceuticals). The protocol was approved by institutional review boards or ethics committees at all participating centers. All patients provided written informed consent.

An independent data and safety monitoring board reviewed unblinded safety data at prespecified intervals and had authority to recommend trial modification or termination. An independent clinical endpoint adjudication committee, whose members were unaware of trial-group assignments, adjudicated all primary and secondary endpoints using prespecified definitions.

2.2 Patient Population

Eligible patients were adults ($\geq$18 years of age) with documented atherosclerotic cardiovascular disease, defined as one or more of the following: (1) prior myocardial infarction ($>$30 days before randomization), (2) prior ischemic stroke ($>$90 days before randomization), or (3) symptomatic peripheral arterial disease with objective evidence of atherosclerotic stenosis.

In addition to documented ASCVD, patients were required to have elevated residual cardiovascular risk, defined as either: (1) 10-year ASCVD risk score $\geq$15% as calculated by the Pooled Cohort Equations, or (2) at least two of the following high-risk features: diabetes mellitus with evidence of target organ damage, chronic kidney disease (eGFR 30 to 59 ml/min/1.73 m^2), or recurrent cardiovascular events ($\geq$2 qualifying ASCVD events in the 24 months before screening).

All patients were required to be on stable GDMT for at least 30 days before randomization, including: (1) antiplatelet therapy (aspirin 75-100 mg daily or clopidogrel 75 mg daily), (2) high-intensity statin therapy or maximally tolerated statin dose with LDL cholesterol $<$100 mg/dL, and (3) blood pressure control (systolic blood pressure $<$140 mm Hg if tolerated).

Key exclusion criteria included: planned coronary, cerebrovascular, or peripheral artery revascularization within 3 months; active bleeding or bleeding diathesis; history of intracranial hemorrhage; severe heart failure (NYHA class IV); severe renal impairment (eGFR $<$30 ml/min/1.73 m^2); or need for oral anticoagulation.

2.3 Randomization and Trial Treatment

After a screening period of up to 14 days, eligible patients were randomly assigned in a 1:1 ratio to receive either cardiovexaban 150 mg once daily or matching placebo. Randomization was performed centrally through an interactive web-response system, stratified by geographic region, index qualifying event, and background antiplatelet therapy.

Patients were seen at clinic visits at 1 month, 3 months, and every 3 months thereafter. Adherence to trial medication was assessed by pill counts at each visit.

2.4 Endpoints

The primary efficacy endpoint was time to first occurrence of the four-point MACCE composite: cardiovascular death, myocardial infarction, stroke, or urgent coronary revascularization. Cardiovascular death included death due to myocardial infarction, heart failure, stroke, sudden cardiac death, or other cardiovascular cause. Myocardial infarction was defined according to the Fourth Universal Definition. Stroke was defined as acute focal neurological deficit lasting $\geq$24 hours or associated with imaging evidence of new cerebral infarction or hemorrhage.

Prespecified secondary efficacy endpoints included: time to first occurrence of the three-point MACE composite (cardiovascular death, myocardial infarction, or stroke); individual components of the primary endpoint; all-cause mortality; and time to first occurrence of cardiovascular death or myocardial infarction.

The primary safety endpoint was TIMI (Thrombolysis in Myocardial Infarction) major bleeding, defined as intracranial hemorrhage (any), clinically overt bleeding with a decrease in hemoglobin $\geq$5 g/dL, or fatal bleeding.

2.5 Statistical Analysis

The trial was designed as an event-driven study powered to detect a 20% relative risk reduction in the primary endpoint with 90% power at a two-sided alpha of 0.05. Based on an assumed annual event rate of 5.5% in the placebo group, we estimated that 1,120 primary endpoint events would be required.

Efficacy and safety endpoints were analyzed in the intention-to-treat population. Time-to-event analyses were performed using Cox proportional-hazards models stratified by randomization factors, with hazard ratios and 95% confidence intervals calculated. Cumulative incidence curves were estimated using the Kaplan-Meier method.

3 Results

3.1 Patients

Between March 2021 and November 2022, 8,206 patients were enrolled and randomly assigned to receive cardiovexaban (4,103 patients) or placebo (4,103 patients). Baseline characteristics were well balanced between groups (Table 1). The mean age was 66.4 years, 28.7% were women, and 78.3% were white. The median 10-year ASCVD risk score was 22.3% (interquartile range, 17.1 to 29.8), and 64.2% of patients had a risk score $\geq$20%. Diabetes mellitus was present in 62.8%, chronic kidney disease in 41.3%, and prior coronary revascularization in 71.4%.

The median duration of follow-up was 36.0 months (interquartile range, 30.2 to 40.1 months). Adherence to trial medication at 36 months was 89.2% in the cardiovexaban group and 90.1% in the placebo group.

3.2 Primary Efficacy Endpoint

The primary endpoint of four-point MACCE occurred in 489 patients (11.9%) in the cardiovexaban group and 643 patients (15.7%) in the placebo group over the 36-month median follow-up (hazard ratio, 0.75; 95% CI, 0.64 to 0.88; $P<0.001$) (Table 2). This corresponds to an absolute risk reduction of 3.8 percentage points and a number needed to treat of 26 patients for 36 months to prevent one MACCE event.

3.3 Secondary Efficacy Endpoints

The three-point MACE endpoint (cardiovascular death, myocardial infarction, or stroke) occurred in 412 patients (10.0%) in the cardiovexaban group versus 531 patients (12.9%) in the placebo group (hazard ratio, 0.77; 95% CI, 0.65 to 0.91; P=0.002).

Individual components of the primary endpoint showed directionally consistent benefit: cardiovascular death occurred in 142 patients (3.5%) versus 181 patients (4.4%) (hazard ratio, 0.78; 95% CI, 0.61 to 1.01; P=0.06); myocardial infarction in 198 patients (4.8%) versus 268 patients (6.5%) (hazard ratio, 0.73; 95% CI, 0.59 to 0.90; P=0.003); stroke in 108 patients (2.6%) versus 157 patients (3.8%) (hazard ratio, 0.68; 95% CI, 0.52 to 0.89; P=0.005); and urgent coronary revascularization in 89 patients (2.2%) versus 109 patients (2.7%) (hazard ratio, 0.81; 95% CI, 0.65 to 1.02; P=0.07).

3.4 Safety Endpoints

TIMI major bleeding occurred in 156 patients (3.8%) in the cardiovexaban group versus 86 patients (2.1%) in the placebo group (hazard ratio, 1.82; 95% CI, 1.39 to 2.38; $P<0.001$) (Table 3). This represents an absolute increase of 1.7 percentage points and a number needed to harm of 59 patients treated for 36 months.

Most major bleeding events were gastrointestinal (109 events [70%] in the cardiovexaban group vs. 58 events [67%] in the placebo group). Intracranial hemorrhage occurred in 16 patients (0.4%) receiving cardiovexaban and 13 patients (0.3%) receiving placebo (hazard ratio, 1.23; 95% CI, 0.59 to 2.58; P=0.58).

4 Discussion

In this large, randomized, placebo-controlled trial involving high-risk patients with established atherosclerotic cardiovascular disease on guideline-directed medical therapy, the addition of cardiovexaban reduced the risk of major adverse cardiovascular and cerebrovascular events by 25% compared with placebo over a median follow-up of 36 months. This benefit was driven by consistent reductions across all components of the composite endpoint, with particularly robust effects on myocardial infarction (27% reduction) and stroke (32% reduction). The efficacy benefit came at the cost of increased major bleeding, predominantly gastrointestinal, with intracranial hemorrhage rates remaining low and balanced between groups.

The results of SHIELD-1 address a critical unmet need in contemporary cardiovascular medicine. Despite widespread use of high-intensity statins, antiplatelet agents, and blood pressure control, patients with established ASCVD continue to experience recurrent events at rates of 4% to 6% annually. The magnitude of benefit observed with cardiovexaban (hazard ratio, 0.75 for the primary endpoint) compares favorably to other secondary prevention strategies tested in recent years.

The observed increase in major bleeding with cardiovexaban is consistent with the known pharmacology of dual-pathway platelet inhibition. The absolute increase of 1.7 percentage points over 36 months (number needed to harm = 59) must be weighed against the absolute MACCE reduction of 3.8 percentage points (number needed to treat = 26). Critically, the excess bleeding risk was driven primarily by gastrointestinal events, which are typically manageable. In contrast, intracranial hemorrhage remained rare and balanced between groups (0.4% vs. 0.3%).

The net clinical benefit analysis, incorporating both ischemic and bleeding outcomes, favored cardiovexaban, suggesting that for every two major bleeding events caused, approximately four MACCE events are prevented. This 2:1 benefit-to-harm ratio supports the use of cardiovexaban in appropriately selected high-risk populations.

Several limitations merit discussion. First, the trial population was predominantly white (78%) and male (71%). Second, background antiplatelet therapy was limited to aspirin or clopidogrel. Third, the trial excluded patients with severe renal impairment and those requiring oral anticoagulation. Fourth, the median follow-up of 36 months may not capture very long-term benefits or late-emerging safety signals.

In conclusion, among high-risk patients with established atherosclerotic cardiovascular disease on guideline-directed medical therapy, the addition of cardiovexaban significantly reduced the risk of major adverse cardiovascular and cerebrovascular events compared with placebo, with an increase in major bleeding driven primarily by gastrointestinal events.

Table 1: Baseline Characteristics of the Patients

Characteristic	**Cardiovexaban (N=4103)**	**Placebo (N=4103)**
Age — yr	66.3 ± 9.8	66.5 ± 9.7
Female sex — no. (%)	1178 (28.7)	1175 (28.6)
Race — no. (%)		
White	3212 (78.3)	3214 (78.3)
Black	412 (10.0)	407 (9.9)
Asian	338 (8.2)	341 (8.3)
Body-mass index — kg/m^2	29.8 ± 5.4	29.7 ± 5.3
Index qualifying ASCVD event — no. (%)		
Prior myocardial infarction	2388 (58.2)	2380 (58.0)
Prior ischemic stroke	1010 (24.6)	1008 (24.6)
Peripheral arterial disease	705 (17.2)	715 (17.4)
Cardiovascular risk factors — no. (%)		
Diabetes mellitus	2577 (62.8)	2574 (62.7)
Chronic kidney disease (eGFR 30-59)	1695 (41.3)	1698 (41.4)
Hypertension	3690 (89.9)	3685 (89.8)
10-year ASCVD risk score		
Median (IQR) — %	22.4 (17.2-29.9)	22.2 (17.0-29.7)
≥20% — no. (%)	2635 (64.2)	2631 (64.1)
Medications at baseline — no. (%)		
Aspirin	2655 (64.7)	2658 (64.8)
Clopidogrel	1448 (35.3)	1445 (35.2)
Statin therapy	4046 (98.6)	4044 (98.5)
High-intensity statin	3381 (82.4)	3376 (82.3)
Laboratory values at baseline		
LDL cholesterol — mg/dL	78.2 ± 28.4	77.9 ± 28.1
eGFR — ml/min/1.73 m^2	68.4 ± 18.7	68.6 ± 18.5

Table 2: Efficacy Endpoints

Endpoint	**Cardiovexaban (N=4103)**	**Placebo (N=4103)**	**Hazard Ratio (95% CI)**	***P* Value**
Primary endpoint				
MACCE (4-point)	489 (11.9)	643 (15.7)	0.75 (0.64-0.88)	<0.001
Secondary endpoints				
MACE (3-point)	412 (10.0)	531 (12.9)	0.77 (0.65-0.91)	0.002
CV death	142 (3.5)	181 (4.4)	0.78 (0.61-1.01)	0.06
Myocardial infarction	198 (4.8)	268 (6.5)	0.73 (0.59-0.90)	0.003
Stroke	108 (2.6)	157 (3.8)	0.68 (0.52-0.89)	0.005
Urgent revascularization	89 (2.2)	109 (2.7)	0.81 (0.65-1.02)	0.07
All-cause mortality	198 (4.8)	241 (5.9)	0.82 (0.67-1.00)	0.05

Table 3: Safety Endpoints

Endpoint	**Cardiovexaban (N=4103)**	**Placebo (N=4103)**	**Hazard Ratio (95% CI)**	***P* Value**
Primary safety endpoint				
TIMI major bleeding	156 (3.8)	86 (2.1)	1.82 (1.39-2.38)	$<$0.001
Components				
Intracranial hemorrhage	16 (0.4)	13 (0.3)	1.23 (0.59-2.58)	0.58
GI bleeding	109 (2.7)	58 (1.4)	1.89 (1.37-2.61)	$<$0.001
Fatal bleeding	12 (0.3)	8 (0.2)	1.50 (0.62-3.65)	0.38
Other endpoints				
TIMI minor bleeding	287 (7.0)	168 (4.1)	1.72 (1.41-2.09)	$<$0.001
Bleeding requiring transfusion	178 (4.3)	98 (2.4)	1.82 (1.42-2.34)	$<$0.001
